

TrES-5b

R-1038

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_160517 · created 2026-07-30T07:29:12+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457525.811 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.052
Transit depth (Rp/Rs)^2	3.4 +/- 1.91 [%]
Orbital Inclination (inc)	83.54 +/- 2.71
Airmass coefficient 1 (a1)	1.081 +/- 0.057
Airmass coefficient 2 (a2)	-0.041 +/- 0.022
Scatter in the residuals of the lightcurve fit is	5.26 %
Best Comparison Star	#2 - [464, 31]
Optimal Aperture	2.22
Optimal Annulus	7.73
Transit Duration (day)	0.068 +/- 0.022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.184 ± 0.052 vs 0.142 (deviation 0.81σ)
Duration fit vs published	0.068 d vs 0.0758 d (deviation 0.36σ)
Mid-time O–C	1.8 min
Depth SNR	3.5
Residual scatter	5.26 %

Light curve

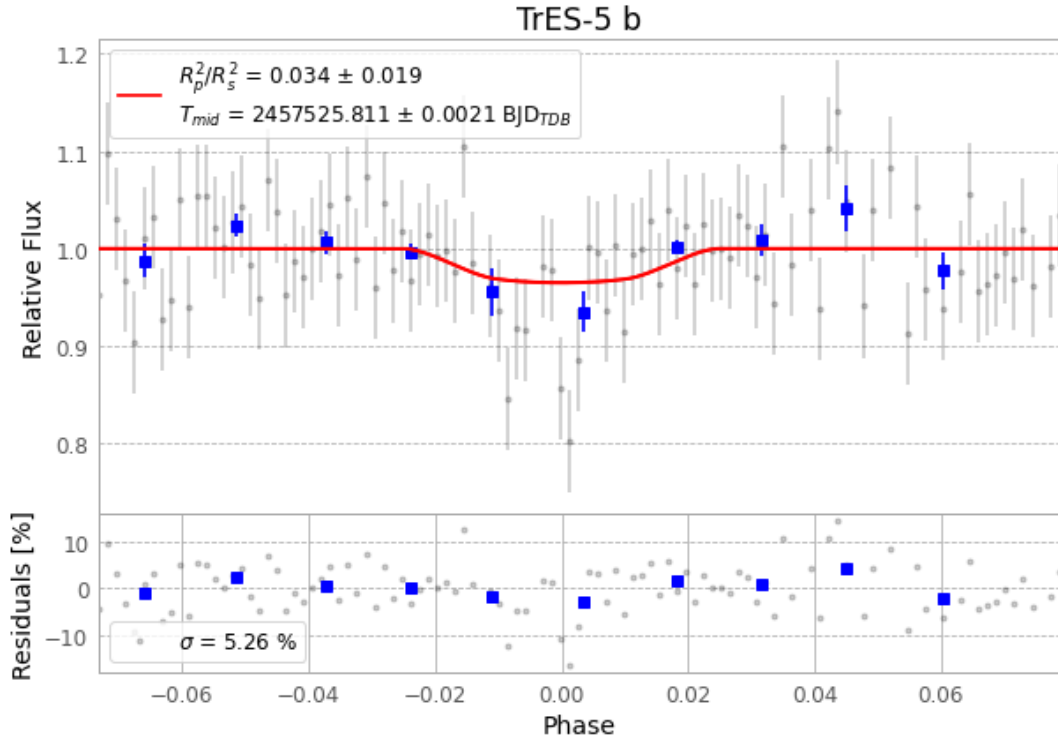


Figure 1 — FinalLightCurve_TrES-5 b_2016-05-16.png (SHA-256 5504b35b2e68c18a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [264, 230], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4c0dc74b58634b6c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 77e40fd

Data provenance

109 FITS frames (72 MB), TRES-5160517045112.FITS → TRES-5160517101512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-160517.log (11f70ffbf062...), FinalLightCurve_TrES-5 b_2016-05-16.png (5504b35b2e68...), FinalLightCurve_TrES-5 b_2016-05-16.csv (030f0cd6bc08...)

Compute resources

Wall time 370.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 07:29Z.